

KATHMANDU UNIVERSITY
University Examination
February/March 2022

Level : BDS

Exam : BDS I

Time : 2 hour 30 minutes

Date : March 15, 2022

Subject: Pharmacology

Paper : 1

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Classify anticholinergic drugs. Write briefly mechanism of action, indications and adverse effects of the prototype anticholinergic drug. [2+1+2+1=6]
2. Define bioavailability. Explain the different factors affecting drug absorption. [1+4=5]
3. Classify antianginal drugs. Write down the mechanism of action of digoxin. [3+2=5]
4. Write short notes on: [3×3=9]
 - a. Beta adrenergic blocking agents
 - b. Therapeutic drug monitoring
 - c. Enalapril 997

SECTION "C"
[25 marks]

5. *Anti-brothel* List fluoroquinolones. Write down the mechanism of action, indications and important adverse effects of fluoroquinolones. [2+2+1+1=6]
6. Classify drugs used in bronchial asthma. Explain the role of inhaled drugs along with its advantages and disadvantages in the treatment of bronchial asthma. [3+2=5]
7. Classify anticoagulants. Write down the mechanism of action and indications of parenteral anticoagulant. [2+2+1=5]
8. Write short notes on: [3×3=9]
 - a. Rationale of combination of sulfamethoxazole and trimethoprim
 - b. Drugs used in Malaria
 - c. Acyclovir

Level : BDS

Paper : I

Exam : BDS I (new course)

Marks : 50

Time : 2 hour 30 minutes

Date : October 31, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. List different routes of drug administration with an example of each. Write two advantages and two disadvantages of sublingual route. [4+2=6]
2. Classify beta blocker with examples. Write the advantages of atenolol over propranolol. [2+3=5]
3. Classify hypolipoproteinemic drugs with examples. Write the mechanism of action and adverse effects of statins. [2+2+1=5]
4. Write short notes on:
a. Calcium channel blockers. [3×3=9]
b. Nomenclatures of drugs with examples.
c. Uses and adverse effects of Atropine.

Antibiotics
bacteriocidal

SECTION "C" [25 marks]

5. Classify cephalosporins with examples. Write its mechanism of action with its common therapeutic uses and adverse effects. [2+2+2=6]
6. Classify drugs used in pulmonary tuberculosis with examples. Write down the mechanism of action of isoniazid with its common adverse effects. [2+2+1=5]
7. Enlist antiplatelet agents. Write down mechanism of action of Aspirin with its two common uses and adverse effects. [1+2+2=5]
8. Write short notes on:
a. Acyclovir. [3×3=9]
b. Drug therapy of iron deficiency and megaloblastic anemia.
c. Bronchodilators.

KATHMANDU UNIVERSITY
August/ September 2021

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : September 6, 2021

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Enumerate the various routes of drug administration with examples. Discuss advantages and disadvantages of sublingual route. [4+2=6]
2. List reversible cholinesterase inhibitors. Write the uses and adverse effects of it. [3+2=5]
3. Classify drugs used in the treatment of heart failure. Write the mechanism of action of digoxin. [3+2=5]
4. Write short notes on: [3×3= 9]
 - a. Therapeutic drug Monitoring.
 - b. Nitrates.
 - c. Adverse drug reaction with examples.

SECTION "C"

[25 marks]

5. Classify penicillins. Write the mechanism of action, uses and adverse effects of penicillin. [2+1+2+1=6]
6. Classify anticoagulants. Write the mechanism of action and uses of oral anticoagulants. (Heparin). [2+2+1=5]
7. Classify drugs used in cough. Write the mechanism of action of bromhexine. [3+2=5]
8. Write short notes on: [3×3= 9]
 - a. Superinfection.
 - b. Chloroquine.
 - c. Bronchodilator.

Level : BDS

Paper : 1

Exam : BDS I

Marks : 50

Time : 2 hr. 30 minutes

Date : April 5, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify anti cholinergic drugs with examples. Write the uses and adverse effects of it. [2+2+2=6]
2. Define drug absorption. Enlist various factors affecting drug absorption. [1+4=5]
3. Classify the drugs used in Angina with examples. Write down the mechanism of action and two common adverse effects of Nitrates. [2+2+1=5]
4. Write short notes on: [3×3=9]
 - a. Mechanism of action of Digoxin
 - b. Propranolol → *formary*
 - c. Biotransformation

SECTION "C" [25 marks]

5. Classify aminoglycosides. Write down the mechanism of action, uses and adverse effects of aminoglycosides. [2+2+1+1=6]
6. Classify antifungal drugs. Write down the mechanism of action and uses of amphotericine B. [2+2+1=5]
7. List the first and second line antitubercular drugs. Write down the common adverse effect of 1st line anti tubercular drugs. [2+3=5]
8. Write short notes on: [3×3=9]
 - a. Streptokinase
 - b. Aminophylline
 - c. Heparin

Level : BDS

Exam : BDS I

Time : 2 hour 30 minutes

Date : March 11, 2020

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"

[25 marks]

1. Classify beta adrenergic blockers with one example. Write down uses, adverse effects and contraindications of it. [3+1+1+1=6]
2. Define Biotransformation. Describe phase I reaction. [1+4=5]
3. Classify the drugs used in the therapy of thromboembolic disorders with one example. Discuss the drugs used in the therapy of megaloblastic anemia with or without neurological involvement. [3+2=5]
4. Write short notes on:
a. Types of antagonist
b. Organophosphate poisoning
c. Mefloquine [3×3= 9]

SECTION "C"

[25 marks]

5. Classify anti-microbial agents on the basis of their mechanism of action with at least one example. Write down the mechanism of action and uses of ciprofloxacin. [3+1+2=6]
6. Classify the drugs used in bronchial asthma. Write down the mechanism of action of methylxanthines. [3+2=5]
7. List the drugs used in the management of myocardial infarction. Mention the difference in the mechanism of action of enalapril, amlodipine and propranolol. [2+1+1+1=5]
8. Write short notes on:
a. Anti-retroviral drugs
b. Digoxin
c. Antihistamines [3×3= 9]

Subject: Pharmacology

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Paper : I

Marks : 50

Date : January 13, 2019

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Enumerate the various routes of drug administration with example. Discuss advantages and disadvantages of oral and parenteral route. [3+3=6]
2. Write down the therapeutic classification of adrenergic drugs with examples. List any two common uses and adverse effects of Adrenaline. [3+2=5]
3. Classify the drugs used in the management of Hypertension with examples. Write down the mechanism of action of Amlodipine. [4+1=5]
4. Write short notes on: [3×3=9]
 - a. Classification of anti-hyperlipidemic drugs with examples
 - b. Biotransformation
 - c. Advantages of Atenolol over Propranolol

SECTION "C" [25 marks]

5. Classify anti-microbial agents on the basis of their mechanism of action with at least one example. Write down the mechanism of action of Ciprofloxacin and mention its two common adverse effects. [3+2+1=6]
6. Classify drugs used in the therapy of Tuberculosis and mention at least one adverse effect of first line drugs. [3+2=5]
7. Write down the difference between Heparin and Warfarin. Write a brief note on uses and adverse effects of Aspirin. [3+2=5]
8. Write short notes on: [3×3=9]
 - a. Antimalarial drugs
 - b. Anti-retroviral drugs
 - c. Bronchodilator

KATHMANDU UNIVERSITY
July/August 2019

CS CamScanner

Level : BDS
Course : BDS I (new course)
Time : 2 hour 30 minutes
Date : August 13, 2019

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

Classify anticholinesterase drugs. Write the mechanism of action, uses and adverse effect of Neostigmine. [3+1+1+1=6]

Enumerate the various routes of drug administration with example. Discuss advantages and disadvantages of intravenous route. [3+1+1=5]

Classify antimalarial drugs. Write the mechanism of action, uses and adverse effect of Chloroquine. [2+1+1+1=5]

Write short notes on: [3×3=9]

- a. Type A adverse drug reaction
- b. Biotransformation
- c. Heparin

SECTION "C"
[25 marks]

Enumerate drugs used in the therapy of Tuberculosis and mention at least one adverse effect of first line drugs. Discuss briefly about the short course chemotherapy. [2+2+2=6]

Classify penicillins. Give the rationale for using the combination of amoxicillin with clavulanic acid. [3+2=5]

Classify drugs used in treatment of congestive heart failure. Write down the mechanism of action of Digoxin. [3+2=5]

Write short notes on: [3×3=9]

- a. Amphotericin B
- b. Bronchodilators
- c. Angiotensin converting enzyme inhibitors

CS CamScanner

KATHIMANDU UNIVERSITY
December 2017/January 2018

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : January 4, 2018

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Enumerate factors affecting drug absorption with suitable examples. Define bioavailability and bioequivalence. Mention the clinical significance of therapeutic half life. [3+2+1=6]
2. Classify drugs used in treatment of angina pectoris. Write mechanism of action, uses and adverse effects of propranolol. [2+1+1+1=5]
3. Classify drugs used in the therapy of congestive heart failure with one example. Write down the mechanism of action of digoxin. [3+2=5]
4. Write short notes on: [3×3=9]
 - a. Treatment of organophosphates poisoning
 - b. Type B adverse drug reactions
 - c. Calcium channel blockers

SECTION "C"
[25 marks]

5. Classify anticoagulant drugs. Describe mechanism of action and therapeutic uses of warfarin. [2+2+2=6]
6. List the classes of drugs used in the therapy of cough with example for each classes. Write down the mechanism of action of bromhexine. [3+2=5]
7. Classify penicillins. Give the basis of use of clavulenic acid with penicillins. [3+2=5]
8. Write short notes on: [3×3=9]
 - a. Primaquine
 - b. Azithromycin
 - c. Anti-tubercular drugs

Course : BDS
Semester : BDS I (new course)
Duration : 2 hour 30 minutes
Date : July 18, 2018

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Define Pharmacokinetics and Pharmacodynamics. Discuss any four major factors affecting drug response. [1+1+4=6]
2. Classify beta-blockers with at least one example of each. List any two common uses and adverse effects of Propranolol. [3+2=5]
3. List the drugs used in the management of congestive heart failure with examples. Write down the mechanism of action of Digoxin. [3+2=5]
4. Write short notes on:
a. Enalapril in hypertension
b. Bioavailability
c. Role of Adrenaline in anaphylactic shock [3×3=9]

SECTION "C" [25 marks]

5. Classify Cephalosporins with at least one example of each. Write down their mechanism of action and list their two common adverse effects. [3+2+1=6]
6. Classify drugs used in bronchial asthma with example. Write down the mechanism of action of Salbutamol. [3+2=5]
7. Classify the drugs used in the therapy of Thromboembolic disorders with one example of each. List the drugs used in the therapy of Megaloblastic anaemia with or without neurological involvement. [3+2=5]
8. Write short notes on:
a. Low dose aspirin
b. Role of Clotrimazole in Oral thrush
c. First line anti-tubercular drugs [3×3=9]

KATHMANDU UNIVERSITY
May/June 2016

Level : BDS

Exam : BDS I (new course)

Time : 2 hr. 30 minutes

Date : June 2, 2016

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify anticholinergic drugs. Mention their therapeutic uses and adverse effects. [2+2+2=6]
2. Define absorption. List factors affecting absorption of drug with at least one example. [2+3=5]
3. List antianginal drugs. Write the mechanism of action and adverse effects of nitrates. [1+2+2=5]
5. Write short notes on:
 - a. First pass metabolism [3×3=9]
 - b. Propranolol
 - c. Classification of antihypertensive drugs

SECTION "C" [25 marks]

6. Classify antitubercular drugs. Write down the category wise treatment of tuberculosis. [2+4=6]
7. Enumerate the anti-malarial drugs. Give the mode of action and toxicities of Mefloquine. [2+2+1=5]
8. Write down the mechanism of action, indications and adverse effects of Gentamicin. [2+2+1=5]
9. Write short notes on:
 - a. Mucolytics [3×3=9]
 - b. Erythropoietin
 - c. Pharmacological basis of using Sulfamethoxazole and Trimethoprim combination

BDS

BDS I

3 hrs.

June 2, 2016

Subject: Pharmacology

Paper : I

Full marks: 70

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

- Classify antimicrobial agents with examples. Write the mechanism of action, uses and adverse effects of penicillin. [3+1+1+1=6]
- Classify β - blockers with examples. Write the mechanism of action, uses and adverse effects of β - blockers. [2+1+1+1=5]
- Enumerate the different routes of drug administration with examples. Write the advantages and disadvantages of intravenous route. [3+1+1=5]
- List anticholinesterase. Write the mechanism of action, uses and adverse effects of anticholinesterase. [2+1+1+1=5]
- List aminoglycosides. Write the mechanism of action, uses and adverse effects of aminoglycosides. [2+1+1+1=5]
- Write short notes on:
 - a. Plasma half life and its importance
 - b. Antagonist
 - c. Tachyphylaxis[3×3=9]

SECTION "B" [35 marks]

- Classify antihypertensive drugs with examples. Write the mechanism of action, uses and adverse effects of captopril. [3+1+1+1=6]
- Classify drugs used in treatment of bronchial asthma. Write the mechanism of action and adverse effects of aminophylline. [3+1+1=5]
- Classify anticoagulants with examples. Write the mechanism of action and uses of oral anticoagulants. [2+2+1=5]
- Classify antimalarial drugs. Write the uses and adverse effects of chloroquine. [3+1+1=5]
- Write the mechanism of action, uses and adverse effects of digoxin. [3+1+1=5]
- Write short notes on:
 - a. Combination therapy in tuberculosis[3×3=9]

BDS

BDS I (new course)

2 hour 30 minutes

October 28, 2016

Subject: Pharmacology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B" [25 marks]

Classify cholinergic receptor agonist. Write down the mechanism of action of pilocarpine and mention its two uses.

[3+2+1=6]

Explain various types of drug antagonism with suitable examples.

[5]

Define pharmacokinetics. Mention various factors affecting drug Absorption.

[2+3=5]

Write briefly on:

a. Digoxin

b. Pralidoxime

c. Captopril - ACE inhibition

[3×3=9]

SECTION "C" [25 marks]

Classify anti-microbial agents on the basis of their mechanism of action with at least one example. Write down the mechanism of action and mention two important adverse effects of amoxicillin.

[3+2+1=6]

List drugs used in cancer. Write down the common adverse effects of anticancer drugs.

[2+3=5]

Classify drugs used in hyperlipidemia with one example. Write down two uses and two adverse effects of aspirin.

[3+2=5]

Write briefly on:

a. Heparin

b. Antihistamines

c. Chloroquine

[3×3=9]

October 28, 2016

Subject: Pharmacology

Paper : I

Marks : 70

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers

Decorate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "A"
[35 marks]

Define bioavailability and bioequivalence. Describe different routes of drug administration with at least one example. Mention advantages and disadvantages of oral route. [2+4+3=9]

Classify antimicrobial agents on the basis of their mechanism of action with one example of each. Write down the mechanism of action of Cloxacillin. Mention four indications and four most common adverse effects of penicillin. [3+2+4=9]

Classify cholinergic receptor agonist with one example. Write down the mechanism of organophosphorus poisoning and mention its antidotes. [2+2+1=5]

Write short notes on:

- Therapeutic drug monitoring
- Agonist and antagonist
- Propranolol
- Fluconazole

[3×4=12]

SECTION "B"
[35 marks]

Classify drugs used in the therapy of bronchial asthma with at least one example. Write the mechanism of action of Theophylline. Mention its two indications with two adverse effects and two drug interactions. [3+3+3=9]

Classify antihypertensive drugs with one example. Write down the mechanism of action of nitroglycerin. List two indications and four adverse effects. [3+3+3=9]

List anticoagulants. Write the mechanism of action of aspirin as an anticoagulant with its two adverse effects. [2+3=5]

[3×4=12]

Write short notes on:

- Antihistamine
- Dyslipidemic drugs
- Anti-malarial drug
- Therapy of Megaloblastic anemia

KATHMANDU UNIVERSITY
2015

Subject: Pharmacology

Level : BDS

Paper : I

Exam : BDS I (new course)

Time : 2 hr, 30 minutes

Marks : 50

Date : September 21, 2015

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Classify cholinergic drugs with examples. Write the mechanism of action, uses and adverse effects of anticholinesterases. [3+1+1+1=6]
2. Define biotransformation. Describe the different phases of biotransformation with examples. [1+4=5]
3. Write the mechanism of action, uses and adverse effects of nitrates. [3+1+1=5]
4. Write short notes on: [3×3=9]
 - a. Enterohepatic circulation
 - b. Propranolol
 - c. Drug dependence

SECTION "C" [25 marks]

5. List the first line antitubercular drugs. Write the mechanism of action, uses and adverse effects of isoniazide. Explain the rational of combination for the treatment of tuberculosis. [1+2+1+1+1=6]
6. Enumerate fluoroquinolones. Write the mechanism of action, uses and adverse effects of it. [2+1+1+1=5]
7. Classify anticoagulants with examples. Write the mechanism of action and uses of heparin. [3+1+1=5]
8. Write the pharmacological basis of : [3×3=9]
 - a. Amoxicillin combined with clavulanic acid in the management of Staphylococcus aureus infection.
 - b. Use of salbutamol in bronchial asthma.
 - c. Primaquine should be given with chloroquine in vivax malaria